SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I KRITHIYA / 192421150 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Krithiya

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks
Concept Identification	20%	1	1	1
Concept Mapping Structure	25%	2	2	2
Linking & Relationships	20%	2	2	2
Application & Analysis	20%	1	1	1
Explanation & Clarity	15%	1	1	1

Total Marks 21 /30

Signature of the Faculty

CONCEPT MAPPING QUESTIONS

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.
Analyze the missing layers and explain how their functions are essential for reliable communication.

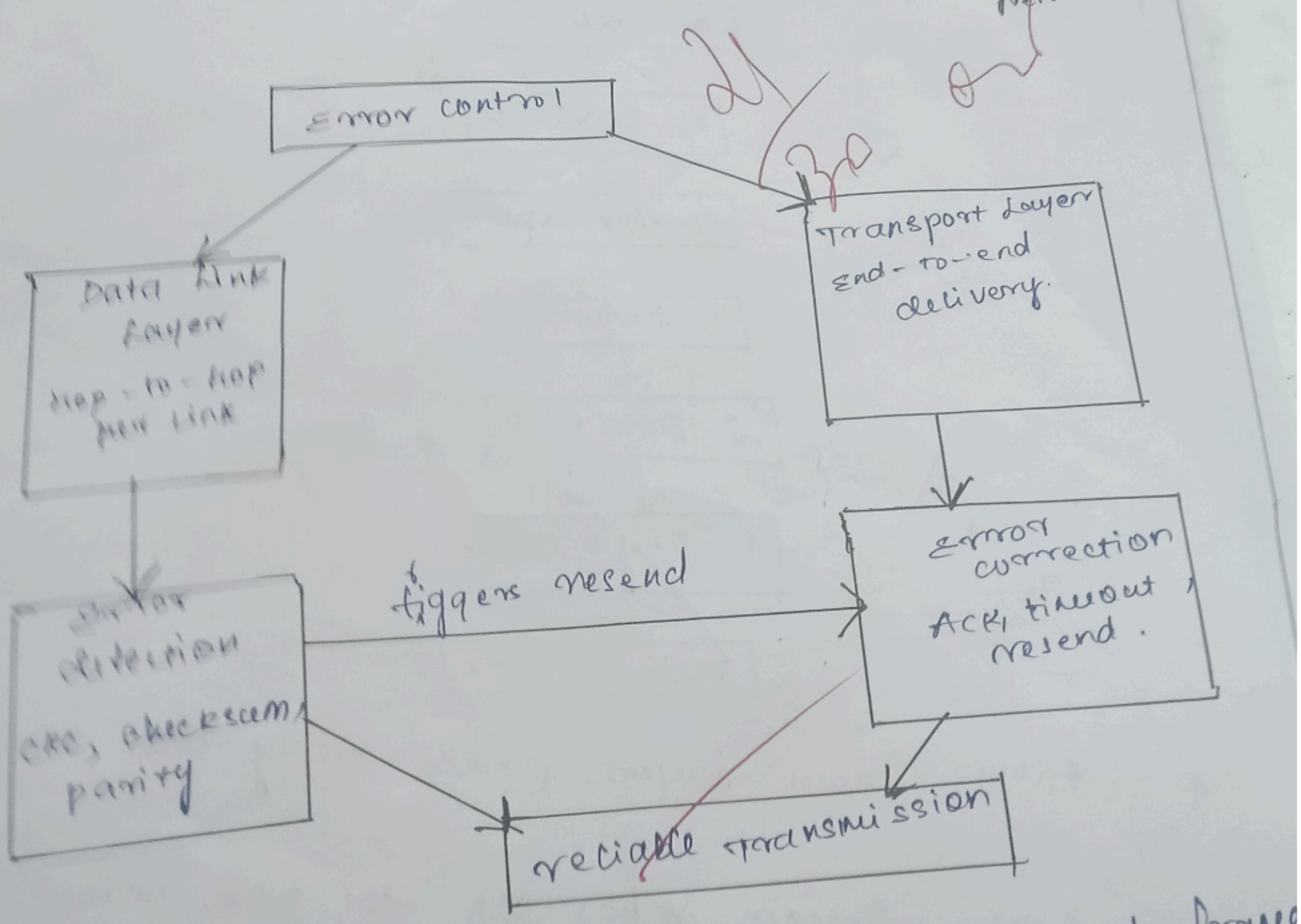
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CSA0710 - Computer Networks

CO1 - AT - 2

Concept MAP

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* Data Link layer : Detects error in each frame using CRC, checksum or parity and discards

* Transport layer : corrects errors using sequence number, ACKs & retransmission

* combined effects :

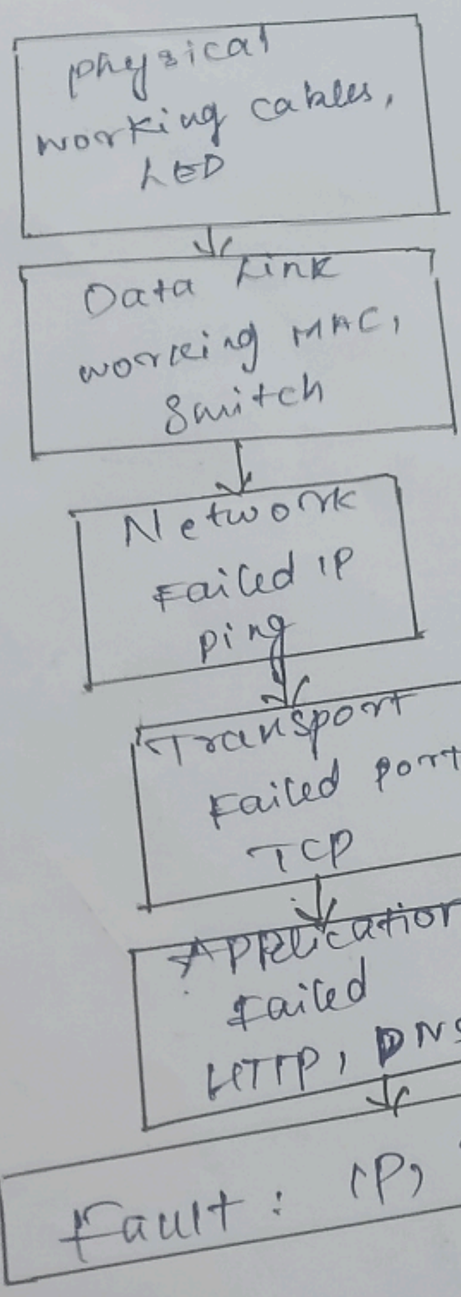
to - end recovery gives ordered. Local detection plus end -

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② 7 Application
↓
6 presentation
↓
5 session
↓
4 transport
↓
3 network
↓
2 Data Link
↓
1 Physical
↓
Web server

* flow down layers 7 to 1 and reply come back up
* Each layer relies on the one below it.
* A fault at any breaks the whole process and the user only sees.

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ASSESSMENT TOOL 3



* problem: The first failure is the Network Layer while layers 1 and 2 work

* upper layers: Transport and Application fail as a result, because they need a working IP path

* It allows bottom-up checking, isolated the faulty layer and points